

AhamX Bodhi

Where I Becomes Infinite

Vision-Language Model Architecture

Module 4.1: Vision-Language Models

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Session Overview

What You Will Learn

- Full VLM architecture component by component
- The critical projection / connector module
- LLaVA, BLIP-2, Flamingo, InternVL designs
- Training stages: pre-training and instruction tuning
- How to select and deploy VLMs in production

Concept at a Glance

- **Duration:** 10 minutes
- **Bloom's Level:** B3 – Apply
- **Difficulty:** L3 – Advanced
- **Module:** 4.1 Vision-Language Models
- **Prerequisites:** ViT (TH-000002), Image Embeddings (TH-000003)



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$1p1 \times \dim y$ $K + 0.100y$



The Three-Component VLM

VLM = Vision Encoder + Connector + Language Model

- **Vision Encoder:** Converts image pixels into patch embeddings. Typically frozen during VLM training. Default: SigLIP-SO/400M or CLIP ViT-L/14
- **Connector / Projection:** The "bridge" that maps vision encoder output dimensions into the LLM's embedding space. This is where most VLM architectural innovation occurs
- **Language Model Backbone:** Autoregressive LLM that processes the combined sequence of visual tokens and text tokens and generates output text. Options: LLaMA-3, Qwen2, Mistral, Phi-3

The connector module is the most critical and variable component. It determines how much visual information is preserved, how many tokens are used, and the training cost.

$$16 \times 16 \times \underline{3} \rightarrow \{R, G, B\}$$
$$224 \times 224$$



Visual tokens are prepended to text tokens in the LLM's context window. The LLM "reads" the image through its token sequence before generating a response.

The Connector: Three Design Choices



$$[E_{img}]_{N \times D_v} = |P| \times d_{img}$$

Connector Architectures Compared

- **MLP Projection (LLaVA-1.5):** 2-layer MLP maps all N patch embeddings from D_v to D_{llm} . Simple, fast, no token compression. Passes all 576–2880 visual tokens to LLM. Best accuracy, highest memory cost.
- **Q-Former (BLIP-2):** A Transformer with N_q learnable query tokens that attend to ViT patch embeddings. Output is N_q fixed tokens (e.g., 32). Strong compression but may lose fine-grained detail. Decouples ViT and LLM training.
- **Resampler / Perceiver (Flamingo, Idefics):** Cross-attention resampler with a fixed number of output tokens. Flamingo uses 64 “media tokens” per image. Efficient for long video (many frames).
- **C-Abstractor (InternVL, LLaVA-HR):** Convolutional downsampling of patch grid before projection. Preserves local spatial structure while reducing token count.

LLaVA-style MLP connectors are currently dominant for single-image tasks; Q-Former/Resampler for multi-image and video.

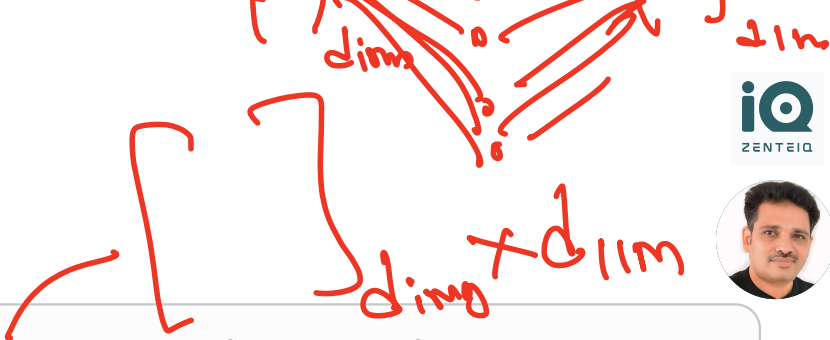


VLM Training Strategy: Two Stages



Standard Two-Stage Training Pipeline

- **Stage 1 – Modality Alignment:** Train ONLY the connector on large-scale image-caption pairs (LAION-CC-SBU, ShareCaptioner data). Freeze ViT and LLM. Goal: align visual and language embedding spaces. 1-2 epochs on ~600k samples. Fast and cheap.
- **Stage 2 – Instruction Tuning:** Fine-tune the connector + LLM (ViT often still frozen) on high-quality visual instruction data (LLaVA-665k, ShareGPT4V). Goal: teach multi-turn conversation, complex reasoning, OCR, chart understanding.
- **Stage 3 (optional) – RLHF/DPO:** Align VLM with human preferences using Direct Preference Optimization on visual instruction preference datasets (RLHF-V, VLFeedback).



Stage 1 is inexpensive (hours on 8 GPUs). Stage 2 quality is determined almost entirely by the instruction dataset quality – curation matters more than scale.





LLaVA Family: The Open Standard

LLaVA Architecture Evolution

- **LLaVA-1.0 (2023)**: CLIP ViT-L/14 + linear projection + LLaMA. First demonstration that MLP connector is sufficient.
- **LLaVA-1.5 (2023)**: CLIP ViT-L/14@336 + 2-layer MLP + Vicuna-13B. State-of-the-art on 11/12 benchmarks at the time.
- **LLaVA-1.6 / LLaVA-NeXT (2024)**: Dynamic resolution tiling, up to 2880 visual tokens. 34B model. Competitive with GPT-4V on many tasks.
- **LLaMA-3.2-Vision (Meta, 2024)**: Official Meta VLM, 11B and 90B. Cross-attention adapter (not MLP projection) fusing ViT features into Llama-3 decoder layers.
- **InternVL2 (2024)**: InternViT-6B encoder + InternLM2/Qwen2 LLM. Achieves 94% of GPT-4V performance at open-weight.

InternVL2-40B is currently the strongest open-weight VLM for document understanding, chart Q&A, and OCR tasks as of early 2025.

> Gum. / Sarvam.



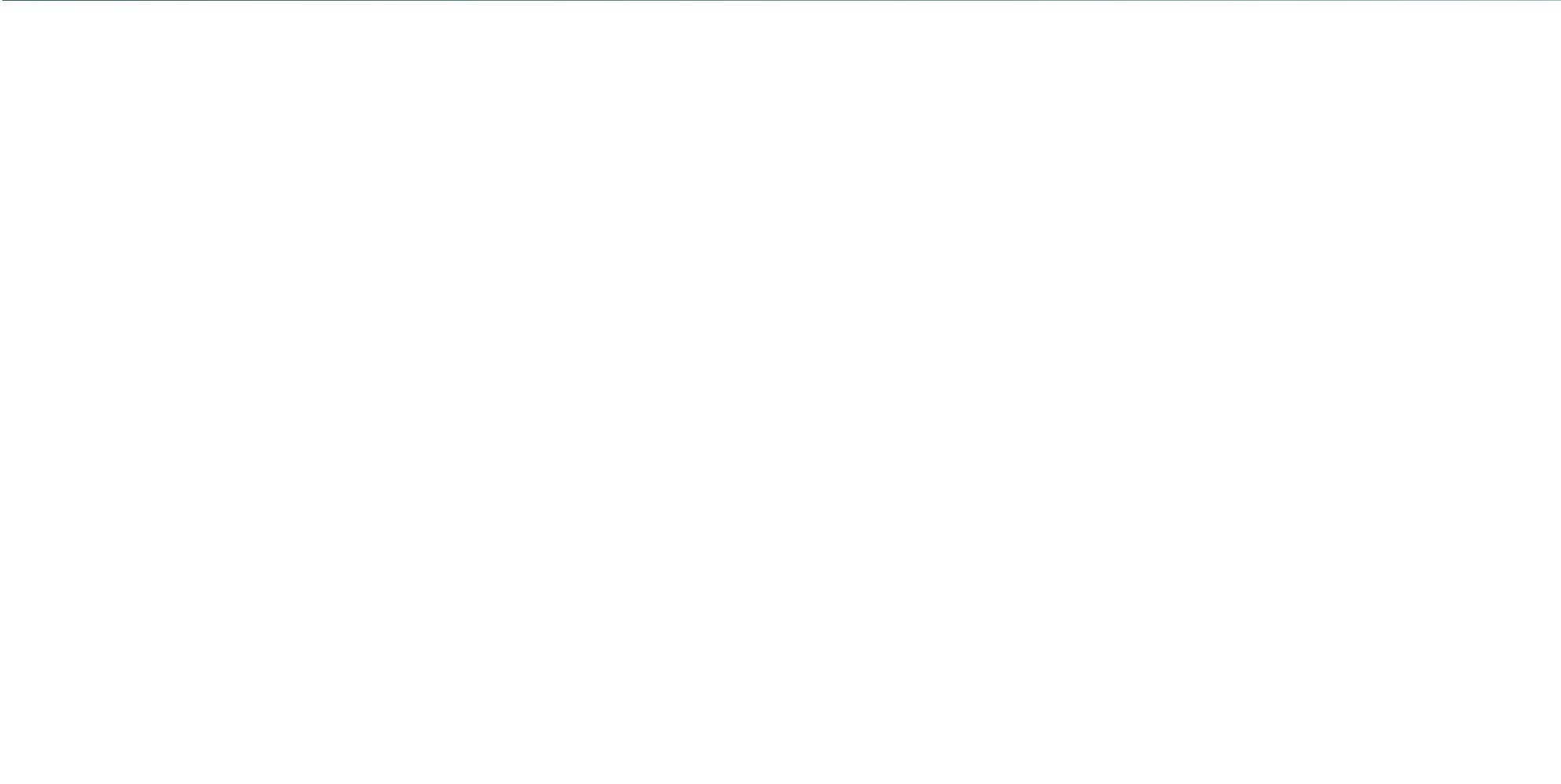
BLIP-2: The Q-Former Approach

BLIP-2 (Li et al., 2023) – Bootstrapped Vision-Language Pre-training

- **Core innovation:** Q-Former is a lightweight Transformer with $N_q = 32$ learnable query tokens that extract the most task-relevant visual information from ViT
- **Stage 1 pre-training:** Train Q-Former with frozen ViT on image-text matching, image-text contrastive, and image-grounded text generation losses simultaneously
- **Stage 2:** Connect Q-Former output (32 tokens) to frozen LLM (FlanT5-XXL or OPT-66B) via linear projection. Only 32 tokens passed to LLM regardless of image resolution.
- **Efficiency:** Decouples ViT and LLM – can swap either without retraining the other. Critical for modular production systems.

Q-Former Trade-off

32 visual tokens vs. 576+ for MLP: faster inference, lower memory, but may miss fine-grained details like small text, dense charts, or spatial relationships.





GenAI Application: VLM API Integration

Building Applications on VLM APIs

- **OpenAI Vision API:** Pass images as base64 or URL in the content array of a chat completion request alongside text. Supports multiple images per message.
- **Anthropic Claude Vision:** image content blocks with `source.type: base64 or url`. Strong document understanding, up to 20 images per request.
- **Local VLM serving:** Ollama supports LLaVA, LLaMA-3.2-Vision locally with a simple REST API compatible with OpenAI SDK.
- **vLLM:** Production-grade serving of LLaVA, InternVL, Qwen-VL with OpenAI-compatible endpoint, PagedAttention, and continuous batching.
- **Cost engineering:** Image processing adds 85–1000 tokens per image depending on resolution and model. High-resolution images are 10–20× more expensive than low-resolution.

Always benchmark your image resolution vs. task accuracy trade-off. For many tasks, 512px is sufficient; only use 1024px+ when fine details are required.



GenAI Application: Visual Quality Control

Manufacturing Defect Detection with VLMs

- **Traditional approach:** Train a CNN classifier on thousands of labeled defect images per product type. Expensive labeling, poor generalization to new defect types.
- **VLM approach:** Describe defect types in natural language prompts. Pass production line images to GPT-4V or InternVL2 with structured output schema (JSON with defect type, location, severity).
- **Zero-shot advantage:** VLMs generalize to new product types without retraining – critical for low-volume manufacturing
- **Pipeline:** Camera → edge preprocessing → VLM API (cloud or local Llava-1.6) → structured JSON → MES/ERP system
- **Real deployment:** Foxconn and BMW reportedly use VLMs for visual inspection of PCBs and automotive body panels

VLM-based inspection is not always more accurate than fine-tuned CNNs, but it is dramatically cheaper and faster to deploy for new product lines.



GenAI Application: Chart and Table Understanding

VLMs for Business Intelligence

- **Task:** Extract data from charts, graphs, and tables in PDFs and presentations – a common enterprise pain point where OCR alone fails
- **State of the art:** InternVL2-40B and GPT-4V achieve $>80\%$ accuracy on ChartQA benchmark; Claude 3.5 Sonnet excels at reading and reasoning over complex tables
- **Output format:** Prompt VLM to return structured JSON with chart type, axis labels, data points, and trend summary
- **Pipeline integration:** PDF $\rightarrow$ page rendering $\rightarrow$ chart detection (DETR) $\rightarrow$ VLM extraction $\rightarrow$ database insert

Engineer's Tip

- Pix2Struct and DePlot (Google) are lightweight models specialized for chart-to-table conversion – faster and cheaper than GPT-4V for this narrow task

VLM Benchmark Landscape (2024--2025)



Model	MMBench	ChartQA	DocVQA	Open?
GPT-4o	83.4	85.7	92.8	No (API)
Gemini 1.5 Pro	82.2	87.2	93.1	No (API)
Claude 3.5 Sonnet	79.7	81.0	89.3	No (API)
InternVL2-40B	82.0	88.4	91.6	Yes
LLaMA-3.2-90V	78.1	83.4	88.2	Yes
LLaVA-1.6-34B	79.3	81.0	74.2	Yes

InternVL2-40B leads on document understanding benchmarks among open-weight models, making it the preferred choice for privacy-sensitive enterprise deployments.

Key Takeaways



What You Should Remember

- VLMs consist of three components: frozen vision encoder (ViT/SigLIP), connector (MLP or Q-Former), and autoregressive LLM backbone
- The connector architecture determines the visual token budget: MLP preserves all tokens (high quality, high cost); Q-Former compresses to fixed count (lower cost, potential detail loss)
- Two-stage training: Stage 1 aligns modalities (train connector only); Stage 2 instruction-tunes the full system on conversation data
- Production: vLLM + InternVL2 for on-premise; OpenAI / Anthropic APIs for managed; Ollama + LLaVA for local development
- High-value applications: document intelligence, chart extraction, visual QA, manufacturing quality control

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